Index Terms: Audio-Visual Speech Separation, Selective Attention, Multi-Scale Fusion, Brain-Inspired Architecture, Cocktail Party Effect
Citations: 44

IIANet: An Intra- and Inter-Modality Attention Network for Audio-Visual Speech Separation

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arXiv: 2308.08143v3 [cs.SD] 2 Feb 2024
International Conference on Machine Learning (ICML), 2024
Code: https://github.com/JusperLee/IIANet

Introduction: The Cocktail Party Problem

Task Definition

- Isolate one voice among many speakers.
- **AOSS (Audio-Only Speech Separation)**
  - Separation quality collapses under severe noise and reverberation.
- **AVSS (Audio-Visual Speech Separation)**
  - Lip motion provides a reliable, speaker-specific visual anchor.

Practical Applications

- Useful for hearing aids, wearable devices, and speech recognition.

Introduction: Research Gaps

Existing Paradigms

- CNNs offer low complexity but fuse at a single temporal scale.
- Transformers use heavy similarity matrices, not designed for speaker selection.

Gap 1: Single Temporal Scale

- Human brain integrates signals at multiple temporal levels.
- Existing AVSS models ignore crucial cross-scale semantic relations.

Gap 2: No Selective Attention

- Intra-modal top-down selective attention is entirely overlooked.

Introduction: The Proposed IIANet

Brain-Inspired Architecture

- Two unimodal networks mimic human auditory and visual pathways.

Variables Definition

- **Mixture audio**: $S \in \mathbb{R}^{1\times T_a}$.
- **Lip frames**: $V \in \mathbb{R}^{H\times W\times T_v}$.
- **Target speech output**: $A \in \mathbb{R}^{1\times T_a}$.

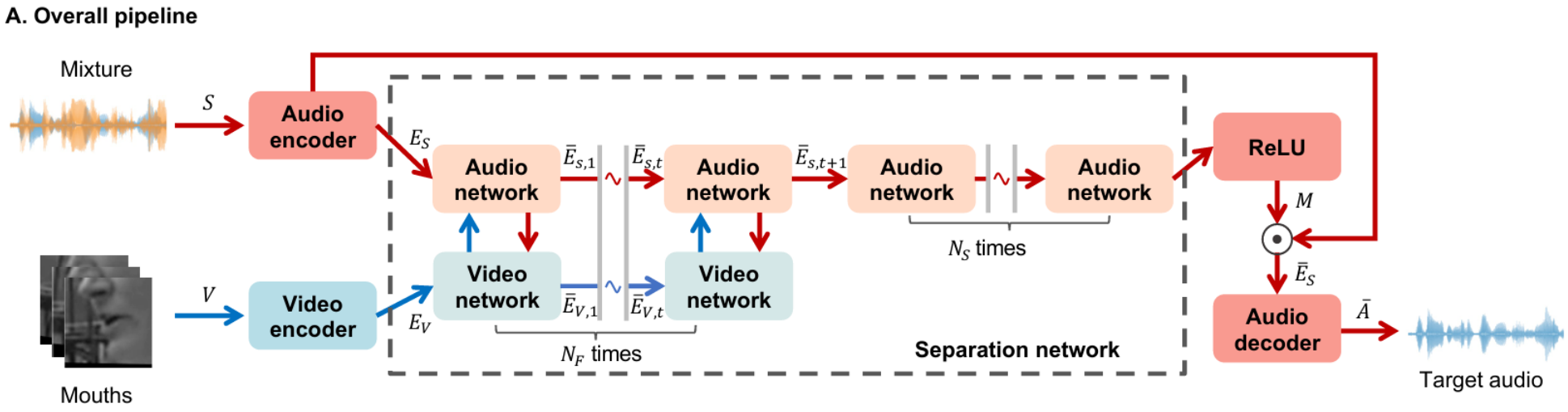
Overall Workflow

- Encoders extract audio and visual embeddings $E_S$and E_V.
- Separation network generates a soft mask M.
- Target is extracted using element-wise multiplication: $\bar{E}_S = E_S \odot M$.

Methodology: Overall Pipeline

Problem Formulation

- **Inputs**: Mixture $S \in \mathbb{R}^{1\times T_a}$, lip frames $V \in \mathbb{R}^{H\times W\times T_v}$.
- **Output**: Target speaker speech $\bar{A} \in \mathbb{R}^{1\times T_a}$.

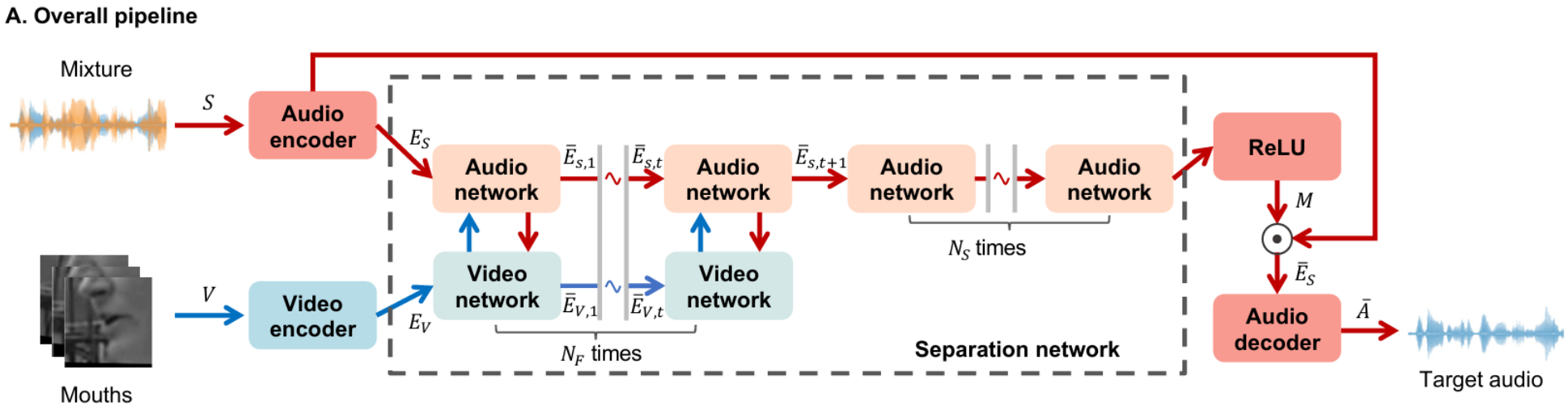


Caption: Fig. 1(A) Li et al. (2024). Overall pipeline of IIANet.

Methodology: Overall Pipeline

Four Modules

- **Audio Encoder**: 1D convolution → E_S.
- **Video Encoder**: Lip-reading pre-trained model (same as CTCNet) → E_V.
- **Separation Network**: Outputs soft mask M via ReLU.
  - $\bar{E}_S = E_S \odot M$.
- **Audio Decoder**: 1D transposed convolution → waveform.

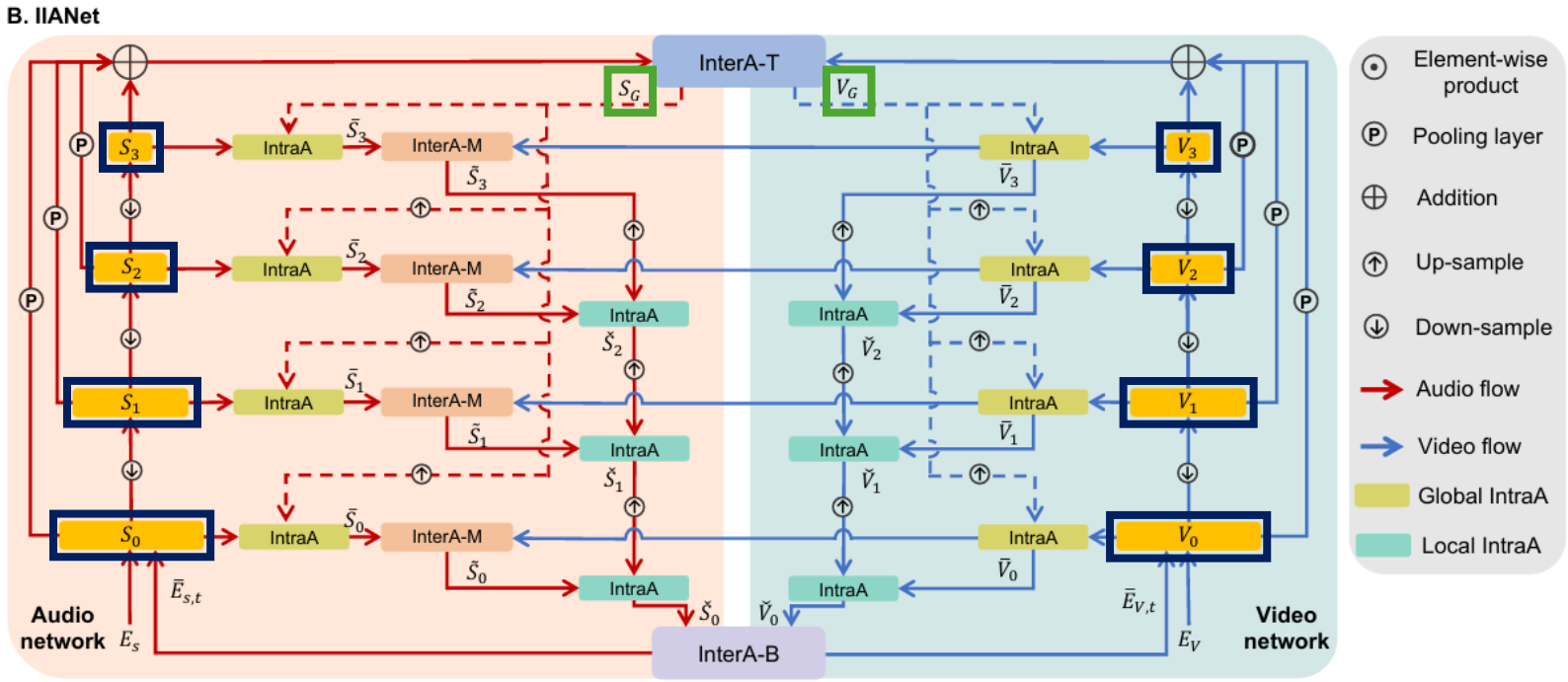


Caption: Fig. 1(A) Li et al. (2024). Overall pipeline of IIANet.

Methodology: Separation Network Workflow

Five-Step Data Flow

1. **Bottom-Up Pass**: Multi-scale features $\{S_i\}, \{V_i\},\ i = 0 \ldots D$
2. **InterA-T**: Pool all scales to coarsest, fuse → global S_G, V_G

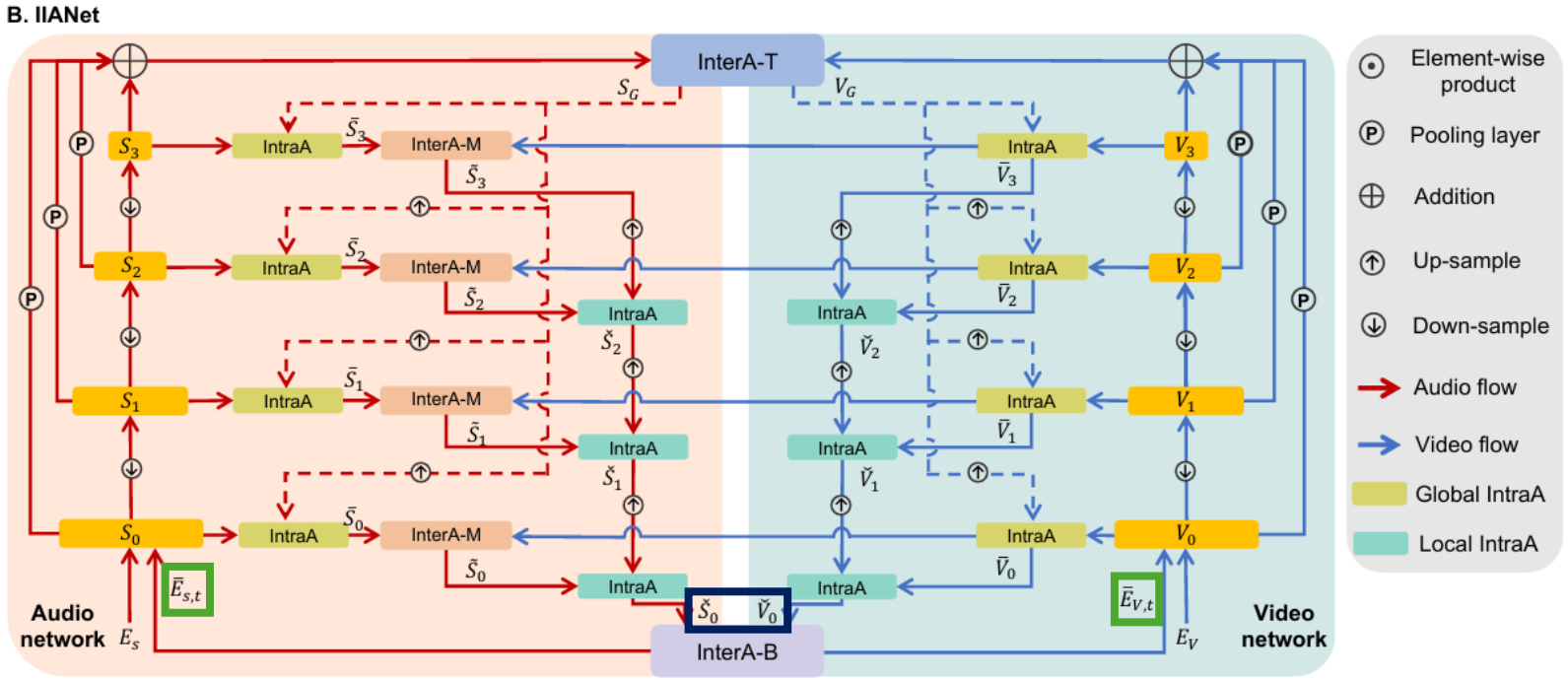


Caption: Fig. 1(B) Li et al. (2024). The separation network of IIANet.

Methodology: Separation Network Workflow

Five-Step Data Flow

3. **Top-Down Pass**: Global IntraA → InterA-M → Local IntraA → $\check{S}_0, \check{V}_0$
4. **InterA-B**: Finest-grained cross-modal fusion → $\bar{E}_{S,t}, \bar{E}_{V,t}$
5. **Cyclic Refinement**
   - N_F AV cycles, then N_S audio-only cycles; weights **shared across cycles**



Caption: Fig. 1(B) Li et al. (2024). The separation network of IIANet.

Methodology: InterA-T (Top Level)

Mathematical Formulation

$$f_S = \sum_{i=0}^{D-1} p_i(S_i) + S_D\,, \qquad f_V = \sum_{i=0}^{D-1} p_i(V_i) + V_D$$

$$S_G = \mathrm{FFN}_S\left(f_S \odot \sigma\big(\mathcal{Q}(f_V)\big)\right), \qquad V_G = \mathrm{FFN}_V\left(f_V \odot \sigma\big(\mathcal{Q}(f_S)\big)\right)$$

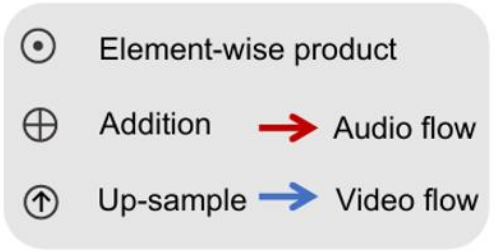
Key Mathematical Components

- **Average Pooling** p_i
  - Scale-specific ratios align all features to the coarsest resolution
- **Gating** $\sigma\big(\mathcal{Q}(\cdot)\big)$
  - One modality generates the attention mask for the other
- **Neuroscience Analogy**
  - Higher associative cortices: frontal and occipital cortex

Methodology: IntraA (Top-Down Attention)

Proposed Form

$$\bar{x} = \phi(x, y) = \sigma\left(\mathcal{Q}(\mu(y))\right) \odot x + \mathcal{Q}(\mu(y))$$



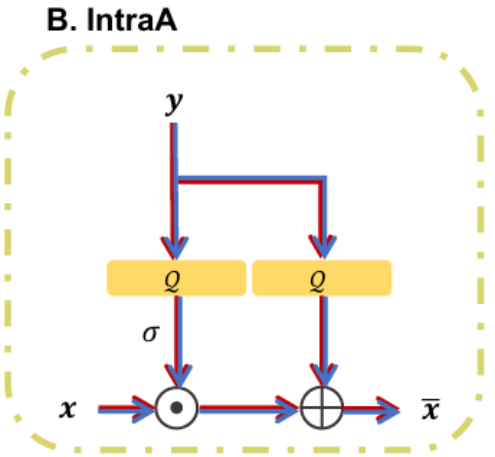
Two Deployments

- **Global IntraA**: $\bar{S}_i = \phi(S_i, S_G), \bar{V}_i = \phi(V_i, V_G)$
- **Local IntraA**: $\check{S}_i = \phi\left(\tilde{S}_i, \check{S}_{i+1}\right)$ — progressive top-down reconstruction

Difference from TDANet

- **Baseline form:** $\phi'(x, y) = \sigma\left(\mu(y)\right) \odot x$ (multiplicative only)
- **Our form:** adds an **additive** high-level injection branch
  - Empirically 16.0 vs 14.7 dB SI-SNRi (Appendix A, Table 6)

Caption: Fig. 2(B) Li et al. (2024). IntraA block flow diagram.

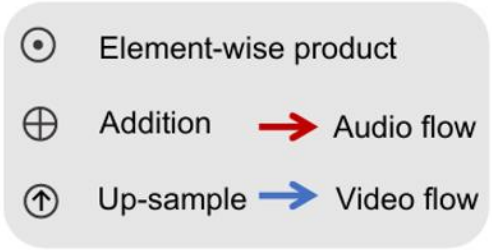


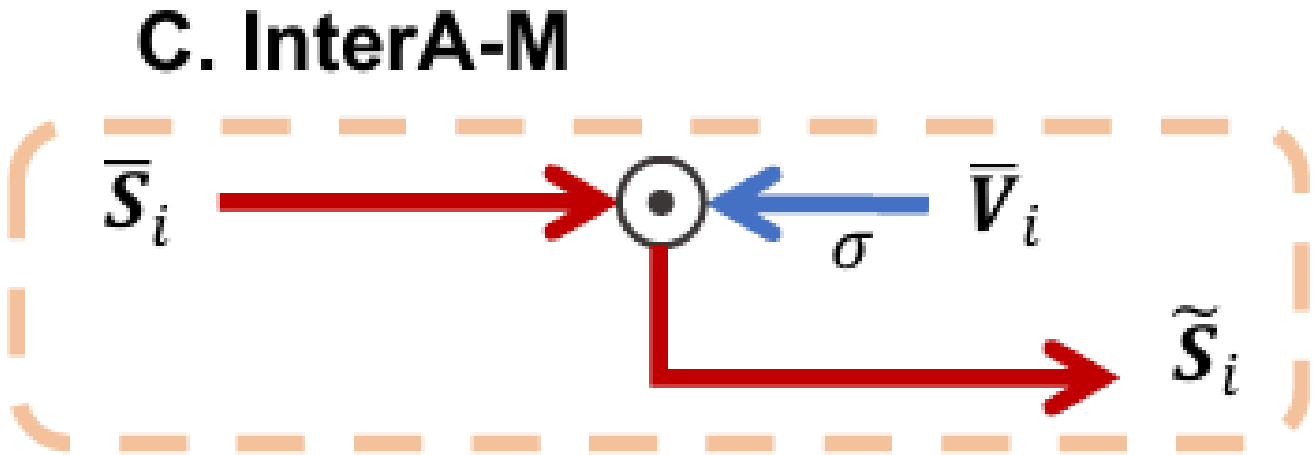
Methodology: InterA-M

InterA-M (Middle Levels)

$$\tilde{S}_i = \sigma\left(\mathcal{Q}\left(\mu(\bar{V}_i)\right)\right) \odot \bar{S}_i \,, \forall i \in [0, \mathrm{D}]$$

- **Unidirectional**: Visual features gate auditory features
- **Applied at every temporal scale**: directly addresses Gap 1
- **Analogy**: Auditory core/belt → V1 direct projections





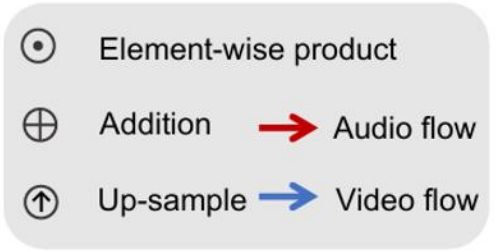
Caption: Fig. 2(C) Li et al. (2024). InterA-M block diagram.

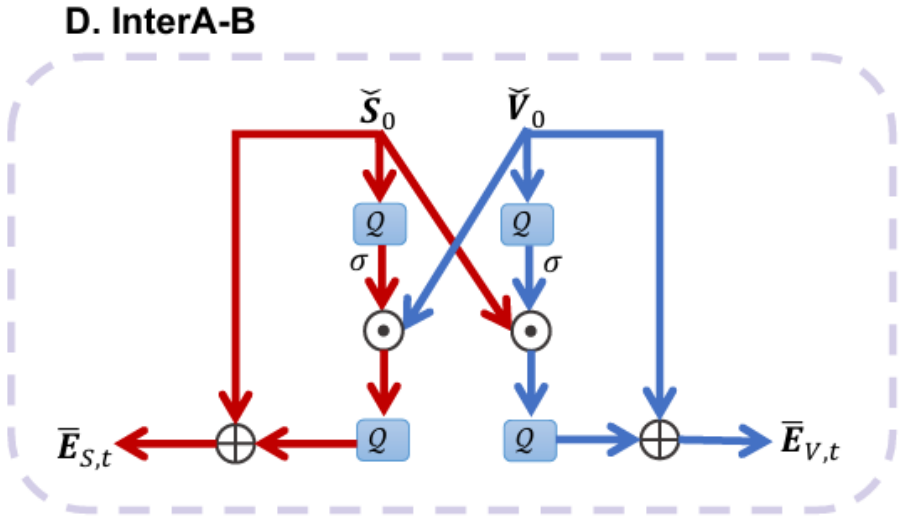
Methodology: InterA-B

InterA-B (Bottom Level)

$$\bar{E}_{S,t} = \check{S}_0 + \mathcal{Q}\left(\mu(\check{V}_0) \odot \sigma\left(\mathcal{Q}(\check{S}_0)\right)\right)$$

- **Bidirectional** fusion at the **finest scale only**
  - CTCNet concatenates all scales → far higher complexity
- **Residual addition:** Prevents over-distortion by the other modality
- **Analogy:** Thalamus





Caption: Fig. 2(D) Li et al. (2024). InterA-B block diagram.

Experiments: Datasets and Setup

Three Benchmarks

- **LRS2 (BBC)**: 11h / 3h / 1.5h — reverberation, partial overlap
- **LRS3 (TED)**: 28h / 3h / 1.5h — relatively clean environment
- **VoxCeleb2 (YouTube)**: 56h / 3h / 1.5h — most complex acoustics
- **Mixtures**: 2 random speakers, SNR $\in [-5,5]$ dB, 2s @ 16 kHz
- **Lips**: 25 FPS, 88×88 grayscale, audio-synchronized

Training Configuration

- **Optimizer**: Adam, lr = 1e-3, ×0.5 on 15-epoch plateau
- **Objective**: SI-SNR (PIT used for multi-speaker settings)
- **Hyperparameters**: $D = 4{,}512$ channels, $N_F = 4$, $N_S = 12$
  - **IIANet-fast**: $N_S = 6$
- **Hardware:** 8 x RTX 4090, batch size 6

Result: Separation Quality

Consistent SOTA Across Benchmarks

- IIANet gains at least **+0.9 dB** SI-SNRi over all baselines
- Gain reaches **+1.7 dB** on complex LRS2 and VoxCeleb2
- IIANet-fast still beats CTCNet by **+1.1 dB** on LRS2

Methods	LRS2			LRS3			VoxCeleb2		
	SI-SNRi	SDRi	PESQ	SI-SNRi	SDRi	PESQ	SI-SNRi	SDRi	PESQ
AVConvTasNet (Wu et al., 2019)	12.5	12.8	2.69	11.2	11.7	2.58	9.2	9.8	2.17
LWTNet (Afouras et al., 2020)	-	10.8	-	-	4.8	-	-	-	-
Visualvoice (Gao & Grauman, 2021)	11.5	11.8	2.78	9.9	10.3	2.13	9.3	10.2	2.45
CaffNet-C (Lee et al., 2021)	-	10.0	1.15	-	9.8	-	-	7.6	-
AVLiT-8 (Martel et al., 2023)	12.8	13.1	2.56	13.5	13.6	2.78	9.4	9.9	2.23
CTCNet (Li et al., 2022)	14.3	14.6	3.08	17.4	17.5	3.24	11.9	13.1	3.00
IIANet (*ours*)	16.0	16.2	3.23	18.3	18.5	3.28	13.6	14.3	3.12
IIANet-fast (*ours*)	15.1	15.4	3.11	17.8	17.9	3.25	12.6	13.6	3.01

Table 1. Separation results of different AVSS methods on LRS2, LRS3, and VoxCeleb2 datasets. These metrics represent the average values for all speakers in each test set, where larger SI-SNRi, SDRi and PESQ values are better. "-" denotes results not reported in the original paper. Optimal and suboptimal performances is highlighted.

Result: Efficiency Trade-off

Hardware-Agnostic Metrics

- vs. CTCNet: 11% MACs, 44% Params, 16% GPU memory
- IIANet-fast: only 7% of CTCNet's MACs

Real Inference Speed

- CPU: 0.52 s vs CTCNet 1.26 s → about 2.4× faster
- GPU: Full IIANet is slower (110 ms vs 84 ms)
  - Many small operators → memory-bound, not compute-bound
- Takeaway: Fewer MACs ≠ automatically faster on GPU

Methods	Computation Cost		Inference Time		GPU Memory (MB)
	MACs (G)	Params (M)	CPU (s)	GPU (ms)	
AVConvTasNet (Wu et al., 2019)	23.8	16.5	1.06	62.51	117.05
Visualvoice (Gao & Grauman, 2021)	9.7	77.8	2.98	110.31	313.74
AVLiT (Martel et al., 2023)	18.2	5.8	0.46	62.51	24.0
CTCNet (Li et al., 2022)	167.1	7.0	1.26	84.17	75.8
IIANet (*ours*)	18.6	3.1	1.27	110.11	12.5
IIANet-fast (*ours*)	11.9	3.1	0.52	70.94	12.5

Table 2. Parameter sizes and computational complexities of different AVSS methods. All results are measured with an input audio length of 1 second, a sample rate of 16 kHz, and a video frame sample rate of 25 FPS. Inference time is measured in the same testing environment without the use of any additional acceleration techniques, such as quantization or pruning.

Result: Multi-Speaker Robustness

Scaling to 3 and 4 Speakers

- Advantage widens as interference increases (+2.3 dB on 3Mix)
- Visual cues become more critical in denser mixtures
- Inference: face detection → lip crop → iterate over each speaker

Methods	LRS2-2Mix		LRS2-3Mix		LRS2-4Mix	
	SI-SNRi	SDRi	SI-SNRi	SDRi	SI-SNRi	SDRi
AVConvTasNet (Wu et al., 2019)	12.5	12.8	8.2	8.8	4.1	4.6
AVLIT-8 (Martel et al., 2023)	12.8	13.1	9.4	9.9	5.0	5.7
CTCNet (Li et al., 2022)	14.3	14.6	10.3	10.8	6.3	6.9
IIANet (*ours*)	16.0	16.2	12.6	13.1	7.8	8.3

Table 3. Performance of different models varies with the number of speakers. These results were based on training conducted using the code from published baseline methods.

Result: Ablation Study

Model	SI-SNRi/SDRi	Params (M)/MACs (G)
Control 1	13.1/13.4	2.2/16.7
Control 2	13.6/14.0	2.2/18.1
IIANet	16.0/16.2	3.1/18.6

Table 4. Importance of IntraA blocks and InterA blocks on the LRS2 test set. MACs are measured with 1s input audio sampled at 16 kHz.

Component Contribution (Table 4)

- Control 1 (no IntraA, no InterA): 13.1 dB
- Control 2 (+IntraA): 13.6 dB → +0.5 dB
- IIANet (+InterA): 16.0 dB → +2.4 dB
  - Cross-modal multi-level fusion is the dominant contributor

InterA Placement (Table 5)

- Single block best: InterA-M alone → 14.5 dB
- Pair best: InterA-M + InterA-B → 15.5 dB
- All three: 16.0 dB — the three levels are complementary

InterA-T	InterA-M	InterA-B	SI-SNRi / SNRi
√	×	×	13.8 / 14.1
×	√	×	14.5 / 14.7
×	×	√	14.1 / 14.4
√	√	×	14.8 / 15.1
√	×	√	15.0 / 15.2
×	√	√	15.5 / 15.7
√	√	√	16.0 / 16.2

Table 5. Impact of each InterA block in IIANet on the LRS2 test set.

Result: Additional Analyses

Visual Cue Analyses

- Visual Contribution (App. F): 16.0 vs 11.2 dB (audio-only) → ≈ +4.8 dB
- Video Encoder (App. J): DC-TCN / MS-TCN / CTCNet-Lip differ < 0.3 dB
  - Lip-reading accuracy does not correlate with separation quality
  - The fusion strategy matters more than the encoder
- Impaired Cues (App. K): Side-profile degradation
  - AVLiT-8 and CTCNet drop > 8%; IIANet only ≈ 4%

Training Analyses

- Fusion Cycles (App. I): NFN F $= 1 \rightarrow 5$ plateaus at 4 (16.0 dB)
- Dynamic Mixing (App. L): LRS2 16.0 → 16.8 dB, only $+0.1$ min/epoch
- Qualitative (App. M): Better high-frequency and harmonic reconstruction

Conclusion

Core Contributions

- **Multi-Level Fusion**
  - Moves cross-modal fusion from a single point to top/middle/bottom
- **Lightweight Selective Attention**
  - Sigmoid gating replaces similarity-matrix cross-attention
- **Efficiency Frontier**
  - New SOTA with 3.1 M parameters and 11% of CTCNet's MACs

Future Directions

- **Long-Form Audio**: Current evaluation is limited to 2-second clips
- **Deployment Optimization**
  - Operator fusion needed to convert MACs savings into GPU speed
- **Broader Multimodal Tasks**
  - Audio-visual segmentation, lip-to-speech, cross-modal medical imaging

My Insights & Critique

Post-Hoc Brain Analogy

- InterA-T/M/B ↔ frontal cortex / cortical projections / thalamus
  - No neural validation experiment; mapping is an analogy, not evidence
  - Stripped of the narrative, the method is multi-scale gated fusion

Reporting Inconsistency

- Abstract claims "40% faster"; Introduction claims "more than twice"
  - Table 2 supports ≈ 2.4× — trust the table, not the abstract

Methods	Computation Cost		Inference Time		GPU Memory (MB)
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My Insights & Critique

Design Asymmetry

- InterA-T and InterA-B are bidirectional, InterA-M is visual → audio only
- No ablation justifies removing the audio → visual path at middle levels

Transferable Value

- Gated multi-scale fusion is cheap enough for 3D medical volumes
  - Directly applicable to CT/MRI synthesis and multi-sequence segmentation
- Replacing sigmoid gating with SSM / Mamba is a natural extension
  - Authors already cite an SSM-based AOSS model (Chen et al., 2023)